

Adversarial Attacks (AA) Review

Robust Deep Learning

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1 Scope

Overall, I think the scope of the report is well chosen and clearly defined. Restricting it to the classical setting of image classification makes the report focused and easy to follow. The only thing I would have found interesting would have been a brief outlook on how attacks differ in other domains (e.g. speech, reinforcement learning, language models, etc.), like in an additional section "Attacks Beyond Image Classification". You even mention prompt injection in the discussion, which felt somewhat abrupt because it had not been introduced earlier in the report. I would also have enjoyed a brief discussion of more recent developments in adversarial machine learning after 2023, even if only to put the surveyed classical attacks into today's broader research context.

The emphasis on evasion attacks appears stronger than on poisoning attacks. If this reflects the literature, mentioning this briefly is maybe helpful.

Besides that, I appreciate that you usually do not only list existing attacks but also discuss their significance, competing explanations for adversarial examples, and their practical relevance.

2 Structure

Introducing the taxonomy before writing about the individual attacks makes the later sections much easier to understand, and Table 1 provides a useful overview.

However, I occasionally found the hierarchy of the attack sections a bit confusing. Some headings introduce specific attacks, others describe broader attack families or concepts. For example, Transfer Attacks seems to mainly motivate the following attacks rather than describing one itself. Clean-Label Attacks is presented separately from Notable Poisoning Attacks although it is itself a type of poisoning attack. A more consistent hierarchy (or a brief explanation of the chosen structure) would make the organization easier to follow. I also felt that the transfer attacks section left some important questions unanswered, such as how an attacker obtains a substitute model and why adversarial examples transfer between different models.

3 Quality of Writing / Clarity of Explanations

I liked that you often explain why certain attacks were important instead of only describing their algorithms. This nicely illustrates the evolution of adversarial machine learning.

In the poisoning section, I found the narrative somewhat weaker than in the evasion section. The evasion attacks are often motivated by explaining why they were introduced and what limitations of previous work they address. The poisoning section, in contrast, mainly describes what each attack does. Personally, I found the earlier narrative more insightful.

I find the motivation for the L_∞ constraint a bit difficult to follow because I was unsure how the statement about sensor inaccuracies relates to the choice of the L_∞ norm. Maybe you could briefly explain that the idea is to model perturbations that remain within the precision of the sensor, instead of simply saying that L_∞ models sensor inaccuracies.

I noticed that some sections are split into many very short paragraphs, which occasionally interrupted the flow of reading. I found this particularly noticeable in the "Jacobian-based dataset augmentation" and "Backdoor poisoning attacks" subsections.

You could further improve the understandability by using a consistent mathematical notation, for example: line 149: target class l , line 180: target class y , line 193: target class y_{true} , line 255: target class t .

Overall, I think the figures are well chosen and helpful for understanding the topic. Figure 4, however, does not seem to be referenced in the running text. I find Figure 5a difficult to interpret. With the text I get an idea of what you are trying to show, but I would consider to improve the clarity. The figures also do not appear to be native LaTeX/vector figures, which means they appear somewhat pixelated, the text is not selectable, and they are not crisp when zoomed in.

I would strongly recommend carefully proofreading the complete report once more before submission. There are numerous small issues throughout the paper that interrupted the reading experience. Since there are quite many, I only list a few examples for each category:

- Typos: where in line 31, Deepfool in line 127, then-s in line 339, successfull in line 477, tha tis in Figure 4a
- Grammar: linearly in line 173, DeepFool is made in line 239
- Missing characters / words / incomplete sentences: lines 177, 190, 197, 217
- Broken references: lines 23, 25
- Citation placeholders: line 174
- Switch between American and British English: optimization and optimisation at various places
- Inconsistent capitalization: line 286

4 Further Comments

- Line 1: The template mentions we should include the abbreviation in the title.
- Table 1: In the table DeepFool is L_2 , but you listed it as an example for L_0 before. I believe the table is correct and the earlier mention should probably be changed.
- Lines 173, 220: At first glance "models are designed to behave mostly linear" and "The highly nonlinear decision boundaries of deep neural networks" seem contradictory. I would consider making the distinction between both concepts a bit clearer.
- Line 308: Missing parenthesis
- Line 330: Setting $\mu = 0$ removes the momentum term, but the method is still iterative. Wouldn't this correspond to BIM?
- Line 485: I was wondering whether CAPTCHA-solving and prompt injection are the best examples here as it seems like those are different concepts than you presented earlier. If the point you wanted to make was that there are no confirmed real-world cases for the presented attacks, I would make that more clear.
- I am not sure if this is incorrect, but I noticed that you placed citations after the period. The seminar template and the papers I read for the seminar place citations before the period. Similarly, I also think a space after words is standard (e.g. after driving in line 11).
- I would recommend using textcite when authors are mentioned in the running text as there are some inconsistencies like in lines 172 or 250ff.
- KGB17b and KGB16 seem to be the same paper.

Overall, I found the report well written and enjoyable to read. My suggestions are mainly intended to improve clarity and consistency before the final submission.